Food Image Recognition and Approximate Calorie Estimation Using Transfer Learning

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Abstract. Many people are advised to track calories and macronutrients, but most apps still depend on manual logging, which is time-consuming and difficult to maintain over the long term. This project investigates a simpler image-based alternative: first recognising the food type in a photo, and then using a nutritional table to assign typical per-dish values for energy, protein, fat and carbohydrates. A ResNet50 model pretrained on ImageNet is fine-tuned on the Food-101 dataset and reaches a test accuracy of 0.7997, which is sufficient to support approximate dish-level calorie estimation. The results indicate that such a system can provide fast approximate feedback, but still struggles with visually similar dishes and does not account for portion size or mixed meals.

Keywords: food image recognition, transfer learning, convolutional neural networks, calorie estimation, Food-101.

1 Introduction

In recent years, calorie and macronutrient tracking has become increasingly popular as healthy eating and conscious consumption have gained attention. Many people are advised to monitor their daily intake of energy, protein, fat, and carbohydrates, both for general well-being and for medical reasons such as weight loss, weight gain, or chronic disease management. Although numerous mobile applications are available to support this process, most of them still rely on manual input of foods and their weights, which is time-consuming, inconvenient, and prone to error. As a result, long-term adherence to manual calorie tracking is often poor.

To reduce this burden, a new generation of applications has begun to use computer vision to estimate nutritional information directly from images. In such systems, users are asked to take one or several photos of their meals, and the application attempts to recognise the food and estimate calories and macronutrients automatically. However, current "food recognition with weight estimation" solutions remain limited in accuracy, especially for mixed dishes and free-living conditions. Portion sizes are difficult to infer from a single image, foods with similar visual appearance are easily confused, and model predictions can vary considerably across lighting conditions and camera viewpoints. Therefore, there

is still a gap between the promise of fully automatic calorie estimation and what is reliably available in practice.

In this project, a more restricted and feasible version of the problem is studied. Instead of estimating exact portion sizes, the focus is placed on robust food type recognition and approximate calorie estimation at the dish level. A deep learning model based on a pretrained convolutional neural network (CNN) is adapted to the Food-101 dataset using transfer learning, so that visual features learned from large generic image collections can be reused for food classification. Once a dish label is predicted, it is mapped to a nutritional database containing typical energy and macronutrient values for that food. Unlike some commercial systems, this project does not attempt to estimate portion size or exact weight; instead, it relies on typical per-dish values to provide a fast, approximate estimate.

This project addresses the following research question: "Can a transfer-learning-based convolutional neural network, trained on Food-101, recognise food images accurately enough to support approximate calorie and macronutrient estimation at the dish level?" The main contribution of this work is a simple end-to-end pipeline that combines food image classification with a nutritional database to provide approximate calorie and macronutrient estimates, together with an empirical evaluation of its performance and limitations on the Food-101 dataset. Model performance is assessed using standard classification metrics on held-out test data, and qualitative examples are used to illustrate how classification errors propagate into calorie estimates. The next section provides background on food image recognition, transfer learning, and related approaches to image-based calorie estimation, in order to place the proposed method in its proper context.

2 Related Work

Food image recognition has been studied extensively, moving from hand-crafted visual features to deep convolutional neural networks. One of the most influential benchmarks in this area is the Food-101 dataset, which contains 101 food categories and 101,000 images and was introduced together with a method that mines discriminative parts for dish recognition using Random Forests [3]. This dataset is widely used for training and evaluating food classification models and is also adopted in the present work.

Transfer learning has been proposed as a way to reuse knowledge from large source domains to improve learning in related target tasks, reducing both data and computational requirements [2]. In computer vision, deep residual networks (ResNets) have become standard backbones for transfer learning, as their residual connections make very deep convolutional networks easier to optimise and yield state-of-the-art performance on ImageNet classification [5]. In this project, a pretrained ResNet-based model is fine-tuned on Food-101 to perform food classification, rather than training a network from scratch.

Several systems have attempted to move from food recognition to calorie estimation. The Im2Calories system demonstrated that a meal could be analysed from one or more images to predict its nutritional contents by combining food recognition, segmentation, and geometric reasoning for volume estimation [4]. More recently, a scoping review of calorie-counting applications for patients with diabetes reported that image-based methods, including those using segmentation, classification, and volume estimation, show promise but still face challenges in accuracy, robustness, and validation in real-world conditions [6]. In contrast, the present work focuses on dish-level classification with approximate calorie and macronutrient lookup, trading fine-grained portion estimation for a simpler and more robust pipeline.

2.1 What did you do differently in the project

In this project, the calorie-estimation task was intentionally simplified by separating it into two independent stages: food classification and nutritional lookup, rather than attempting full portion-size estimation or multi-item segmentation, which remains technically challenging and inaccurate in current systems such as Im2Calories [4]. A transfer-learning approach was adopted so that a pretrained ResNet model [5] could be adapted to the Food-101 dataset [3] instead of training a deep network from scratch. Two transfer-learning strategies were evaluated, a frozen feature extractor and a partially fine-tuned model, to examine how different levels of adaptation influence classification accuracy. Finally, the predicted food labels were mapped to a nutritional table to provide approximate calorie and macronutrient values, offering a lightweight end-to-end pipeline that prioritises robustness and practicality over precise volume estimation, which remains an open challenge in image-based dietary assessment [6].

3 Methodology

3.1 Data Generation

Fig. 1: Sample images from the Food-101 training set.

The experiments in this project were conducted using the Food-101 dataset, which contains 101 food categories and a total of 101,000 images [3]. The dataset is organised into two components: an `images` directory containing one folder per class, and a `meta` directory providing the official training and test split files.

The metadata files (`classes.txt`, `train.txt`, and `test.txt`) were parsed to generate explicit lists of image paths and numeric labels. Each entry in the split files specifies an image as `class_name/image_id`, which was converted to a full file path and mapped to an integer label in the range of 0–100.

All images were loaded using the Python Imaging Library (PIL) and preprocessed using the `torchvision` library. Images were resized to 224×224 pixels, converted to RGB tensors, and normalised using the ImageNet mean and standard deviation so that they matched the input distribution expected by pretrained convolutional neural networks [1]. A custom PyTorch `Dataset` class was implemented to return a transformed image and its label for any given index. The data pipeline was completed using `DataLoader` objects, enabling efficient shuffling, batching, and multi-threaded loading during training and evaluation.

3.2 Modelling Process

A convolutional neural network (CNN) based on the ResNet architecture was adopted as the primary model. Residual networks are widely used in transfer learning due to their stable optimisation behaviour and strong generalisation performance on large-scale datasets such as ImageNet [5]. In this work, a ResNet50 model pretrained on ImageNet was loaded from the `torchvision` model repository and adapted to the Food-101 classification task.

Transfer learning was applied to reuse the high-level visual representations learned from natural image data, reducing both the computational cost and the amount of task-specific data required for effective training [2]. The final fully connected layer of the pretrained model was replaced with a new classification head containing 101 output units. Two training strategies were explored. In the baseline configuration, all pretrained layers were frozen and only the classification head was trained, treating the model as a fixed feature extractor. In the second configuration, the final residual block and the classification head were unfrozen and fine-tuned with a lower learning rate, allowing the network to adapt high-level features more closely to the domain of food imagery.

The model was trained using mini-batch gradient descent with cross-entropy loss, following standard practice for multi-class image classification. A portion of the training data was set aside for validation to monitor generalisation performance, while the held-out test set provided the final accuracy score. After classification, predicted food labels were mapped to a small nutritional lookup table to estimate calories and macronutrient values per serving, following the general idea of combining visual food recognition with nutritional retrieval [4, 6]. The nutritional table was constructed from typical per-serving values for each Food-101 class, based on standard food composition information.

4 Experimentation and Results

4.1 Experimental Setup

All experiments were carried out in Python using PyTorch on Google Colab, with an NVIDIA T4 GPU provided by the Colab runtime. The Food-101 dataset was

obtained from its official release [3] via a public Kaggle mirror, and stored in the Colab environment for training and evaluation.

The official training split of 75,750 images was further divided into a training and validation subset using a 90/10 random split, yielding 68,175 images for parameter updates and 7,575 images for validation. The official test split of 25,250 images was kept separate and used only for final evaluation. A fixed batch size of 32 was used throughout.

Both the baseline and fine-tuned models were trained using the same optimisation setup, except for the learning rate and the set of trainable parameters. Cross-entropy loss was employed as the objective function. For the baseline model, only the final fully connected classification layer was trainable, and the Adam optimiser was used with a learning rate of 10^{-3}. A step learning-rate scheduler reduced the learning rate every 3 epochs. The baseline model was trained for 5 epochs.

Model	Test Accuracy	Test Loss
Frozen ResNet50 baseline	0.6214	1.4482
Fine-tuned ResNet50 (layer4)	0.7997	0.7603

Table 1: Comparison of baseline and fine-tuned models on the Food-101 test set.

For the fine-tuning experiment, a new ResNet50 initialised with ImageNet-pretrained weights was configured so that the last residual block (`layer4`) and the final classification layer were trainable, while all earlier layers remained frozen. A smaller learning rate of 10^{-4} was used to prevent destabilising the pretrained weights. The fine-tuned model was trained for 8 epochs using the same train/validation split.

To analyse model behaviour beyond overall accuracy, predictions and true labels were collected for the entire test set. These were used to compute per-class accuracies and identify typical misclassifications. Qualitative examples of both correct and incorrect predictions were also visualised. Finally, the predicted class was mapped to the nutritional table to demonstrate how classification errors propagate into calorie estimates.

4.2 Discussion of Results

The frozen-backbone baseline achieved a test accuracy of 0.6214. Over 5 epochs, training accuracy increased from roughly 0.41 to 0.62, while validation accuracy improved from approximately 0.49 to 0.57. Loss decreased consistently, indicating stable training and good use of the fixed ImageNet features.

Fine-tuning the last residual block led to a clear improvement, with test accuracy rising to 0.7997. Validation accuracy stabilised at around 0.75. This

Fig. 2: Examples of misclassified test images. The true label is shown above each image, and the predicted label is shown below.

shows that adapting the high-level features to food-specific patterns helps the model form sharper class boundaries and generalise better.

A closer look at misclassified examples reveals systematic confusions. Figure 2 shows several apple pie images that were predicted as entirely different dishes, including cheesecake, samosa, onion rings, bread pudding, dumplings, and even savoury foods such as hummus and french onion soup. These mistakes usually occur when the texture or colour resembles another class, or when the plating makes the dish appear visually ambiguous.

Fig. 3: Classified examples with predicted labels and confidence scores.

In contrast, the model performs very well on dishes with highly distinctive visual cues. As shown in Figure 3, categories such as edamame, french fries, panna cotta, hot dog, and waffles are classified correctly with high confidence.

These foods tend to have stable shapes and colours across images, which makes them easier for the network to recognise.

Fig. 4: Example predictions with estimated calories and macronutrients (protein, fat, carbohydrates) per serving.

Finally, Figure 4 illustrates how the classification output was extended to approximate calorie and macronutrient values using a nutritional lookup table. While the estimates are only approximate, since portion size and mixed meals are not modelled, this example demonstrates how even a simple classifier can support automated diet tracking. Errors in classification, however, directly propagate into the calorie estimate, highlighting a key limitation of this approach.

5 Conclusion

This project investigated the use of transfer learning with a ResNet50 model for food image recognition on the Food-101 dataset and showed how the resulting

classifier can be connected to a simple calorie- and macronutrient-estimation component. A baseline model that used the pretrained network as a frozen feature extractor and trained only a new classification head achieved a test accuracy of 0.6214, showing that ImageNet features provide a strong starting point for food classification. By partially fine-tuning the last residual block in addition to the classification head, the test accuracy increased to 0.7997, indicating that adapting high-level features to the food domain leads to substantial gains in performance while keeping the training cost moderate.

Analysis of per-class accuracies and misclassified examples showed that the fine-tuned model performs best on visually distinctive dishes and struggles most with fine-grained distinctions between similar foods, particularly meat dishes and desserts with overlapping visual characteristics. These findings are consistent with known challenges in fine-grained visual recognition and suggest that further improvements may require more specialised architectures or loss functions designed for fine-grained categorisation. The calorie-estimation component illustrated how model predictions can be mapped to approximate nutritional information per serving, but also underlined important limitations related to portion size, composite meals, and the propagation of classification errors.

Future work could focus on making the system more realistic for dietary tracking. Potential directions include multi-label recognition of composite meals, object detection or segmentation to separate different items on a plate, and explicit modelling of portion size, for example through depth cues or volume estimation. Despite these limitations, the results demonstrate that transfer learning with modern CNNs can provide a strong foundation for food recognition and that even a simple nutritional lookup can turn raw predictions into a practically meaningful application.

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